

K1 MUSE Shelf

RISC-V Architecture For Intelligent Future



RISC-V cloud computing power open for global developers

K1 MUSE Shelf is a RISC-V development server designed specifically for the developer community, offering comprehensive computing power, efficient operations, high-speed connectivity, and intelligent management. The application platform is powered by SpacemiT Cloud services, offering a highly available, scalable, and high-performance distributed cloud environment for RISC-V-based application development and adaptation.

K1 MUSE Shelf aims to provide global developers with RISC-V computing instances, supporting features like remote desktop, remote compilation, one-click flashing, and one-click application installation, allowing developers to leverage RISC-V processing power and engage in creative development anytime and anywhere.



High-Density RISC-V Compute Cluster

Integrates 80 SpacemiT RISC-V 8-core processors, delivering massive parallel compute for development and validation



Remote Flashing & System Management

Supports Fastboot flashing and remote power control for efficient device management and system updates



Standard Server Form Factor

Built on a 2U server platform for easy data center integration and large-scale operations



Flexible Debugging Support

Supports SSH, UART, ADB, and other methods across different development stages



Remote Development Capability

Enables file transfer and VNC/RDP access for a complete remote development experience



Modular Blade Architecture

2/4-blade design for flexible deployment, fast maintenance, and scalable expansion



Specifications

Array chassis	BMC	AC50-BMCS-SRV-RISCV-2.1.4
	Cloud services	SpacemiT Cloud
	Chassis	2U rack with 4 blades
	Number of CPU Boards	20 CPU boards per blade, 4 blades total, 80 K1 SoMs
K1-SOM	CPU	K1 (8-core RISC-V CPU with 50K DMIPS performance and 2 TOPS AI processing power)
	DRAM	8GB / 16GB
	Storage	64GB / 128GB
	Interfaces	GbE / USB Device / UART
	Dimension	47.10 mm × 64.10 mm
	PCB	6-layer, through-hole via
Switching system	Network	2 × 25 Gb SFP28 optical ports per blade, 4 blades, 8 optical ports total. Each blade with independent networking
Power	Input	1200 W Platinum power module, hot-swappable, 1 + 1 redundancy
Environment	Temperature	5°C~35°C
	Relative Humidity	40% ~ 60%